

Task 1.1

Proof: $(\Rightarrow) y = f(x) + \varepsilon, \varepsilon \sim N(0, \Sigma)$

We have $\varepsilon = y - f(x)$.

If x is given, $y = \varepsilon + f(x) \sim N(f(x), \Sigma)$.

(By Transformation laws: $Y = BX + c$,

$X \sim N(\mu, \Sigma)$, then $Y \sim N(c + B\mu, B\Sigma B^T)$)

$(\Leftarrow) y = f(x) + \varepsilon, y|x \sim N(f(x), \Sigma)$.

We can show it similarly.

Task 1.2.

$\in \mathbb{R}^{n \times n}$

$y|x = X\beta + \varepsilon \sim N(X\beta, \sigma^2 I_n)$ from the result of 1.1. Thus:

$$p(y|x) = \frac{1}{\sqrt{(2\pi)^n / \sigma^2}} \exp\left(-\frac{\sigma^2}{2} (y - X\beta)^T (y - X\beta)\right)$$

$$\log p(y|x, \beta, \sigma^2) = -\frac{\sigma^2}{2} (y - X\beta)^T (y - X\beta) - \frac{n}{2} \log(2\pi) + \log \sigma. (\#)$$

To minimize $(\#)$ is equivalent to maximize $\|y - X\beta\|_2^2$ as
 $-\frac{\sigma^2}{2}$ is a negative constant factor and
 $-\frac{n}{2} \log(2\pi) + \log \sigma$ is constant.

Task 1.3.

$$\hat{\beta}_{MLE} = \arg\max_{\beta} \|y - X\beta\|_2^2$$

$$\frac{\partial}{\partial \beta} \|y - X\beta\|_2^2 = -2X^T(y - X\beta) = 0 \Rightarrow X^T X \beta = X^T y$$

$$\hat{\beta}_{MLE} = (X^T X)^{-1} X^T y$$

This estimator is unique since X has full column rank. Hence $X^T X$ is invertible.

$X^T X \beta = X^T y$
(normal equation)

To derive the covariance, substitute y by $X\beta + \varepsilon$.

$$\begin{aligned} \hat{\beta}_{MLE} &= (X^T X)^{-1} X^T (X\beta + \varepsilon) \\ &= \beta + (X^T X)^{-1} X^T \varepsilon \\ \text{Cov}(\hat{\beta}_{MLE}) &= \text{Cov}(\beta + (X^T X)^{-1} X^T \varepsilon) \\ &= \text{Cov}((X^T X)^{-1} X^T \varepsilon) \\ &= (X^T X)^{-1} X^T \text{Cov}(\varepsilon) X (X^T X)^{-1} \\ &= (X^T X)^{-1} X^T X (X^T X)^{-1} \text{Cov}(\varepsilon) \\ &= \sigma^2 (X^T X)^{-1} \end{aligned}$$